

Cybersecurity Incident Management

The Cybersecurity Incident Management System is a web-based application designed to help organizations report, track, organize, and manage cybersecurity incidents. The purpose of the software is to provide users with a centralized system where security-related incidents can be documented and monitored from the time they are reported until they are resolved. Some examples of incidents that can be recorded include phishing attacks, malware infections, unauthorized access, suspicious login activity, data breaches, and other security events. The system will provide a simple user interface (UI) that allows users to interact with the application without needing to use command-line tools.

The primary users of the system will be cybersecurity analysts, IT administrators, and authorized employees. Employees will be able to submit incident reports containing information such as the incident type, description, date, severity, and affected system. Cybersecurity analysts and administrators will be able to view submitted incidents, search and filter reports, update incident statuses, and review information needed to investigate security events. The system will use different types of user access so that authorized users can perform administrative functions while regular users have access to the features necessary for submitting and viewing their own reports.

The application will use HTML and CSS to create the user interface. HTML will provide the structure of the web pages, including forms, buttons, navigation menus, tables, and incident-reporting fields. CSS will be used to organize the layout and improve the appearance of the application so that it is easy to navigate. JavaScript will be used to provide interactive functionality on the client side, such as form validation, filtering information, and responding to user actions.

Python with the Flask framework will be used for the backend of the application. Flask is being selected because it is lightweight, relatively easy to develop with, and provides the necessary functionality for creating a web application and connecting the user interface to the backend. Python will process user requests, validate submitted information, communicate with the database, and return the appropriate information to the user. This separation between the frontend and backend will also make the application easier to maintain and expand.

SQLite will be used as the database management system. The database will store information such as user accounts, incident reports, incident types, severity levels, dates, descriptions, and incident statuses. SQLite is appropriate for this project because it is lightweight, does not require a separate database server, and can be easily integrated with Python. The database connection will allow information entered through the UI to be permanently stored and retrieved when needed.

GitHub will be used to store and submit the project source code and documentation. The repository will contain the application's source files, database-related code, README documentation, and project description. GitHub will provide version control and allow changes to the project to be tracked throughout development.

Overall, the Cybersecurity Incident Management System will combine a web-based UI, Python backend, and database connection to create a functional application for managing cybersecurity incidents. The project will demonstrate important software development concepts including user interfaces, backend programming, database management, data validation, and basic cybersecurity practices.